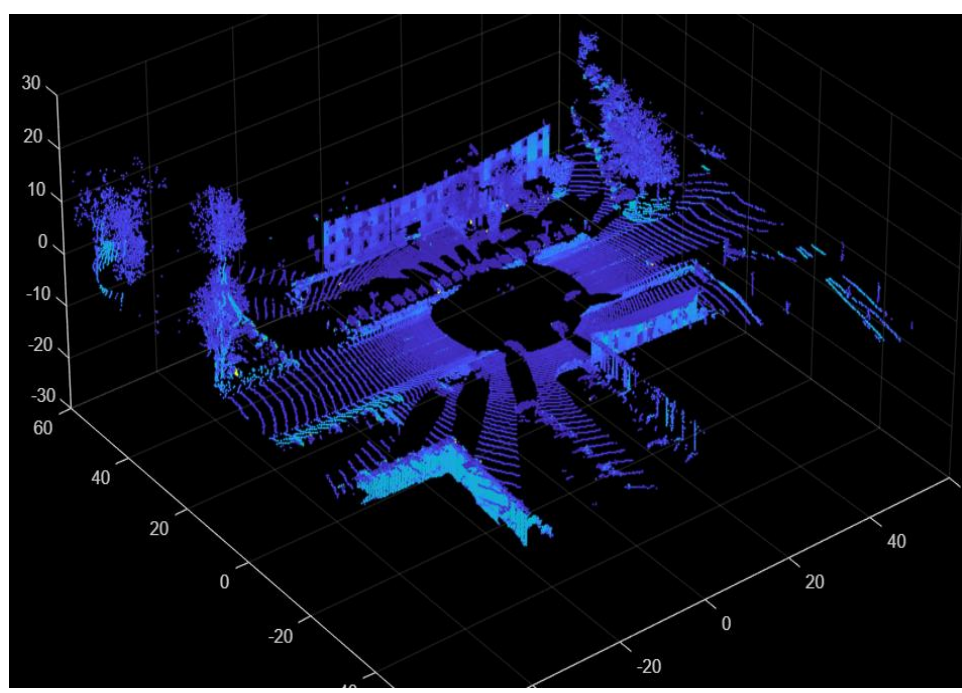
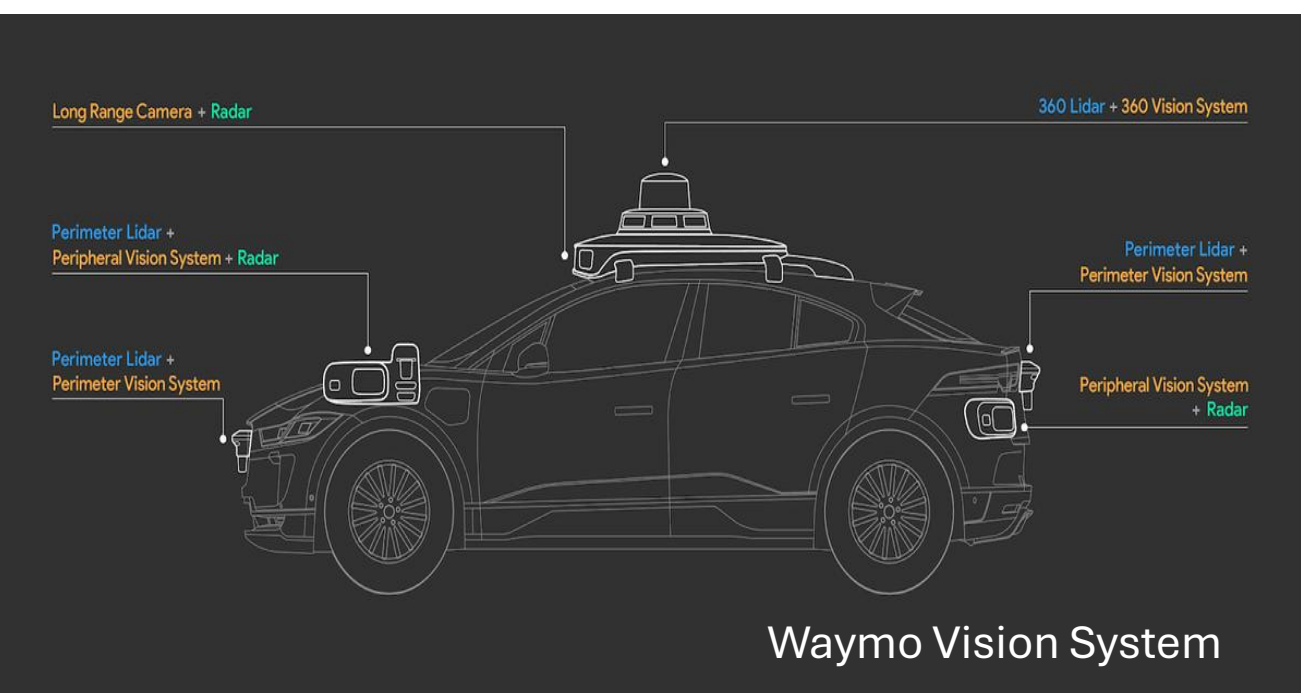


Benchmarking Interpolation Methods for LiDAR Image Reconstruction

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Motivation

Autonomous vehicles are a rapidly developing technology with the potential to improve the safety and efficiency of daily transportation, but require precise sensing to determine position and motion. LiDAR scanners are a critical component, directly measuring surface distances by emitting laser pulses and recording their return time. However, as the vehicle moves, each scan is captured from a different position, making direct frame-to-frame comparison impossible without resampling. The Fourier transform can exactly reconstruct any sampled signal and evaluate it at arbitrary positions motivating an investigation into whether making frequency-domain methods can outperform simpler geometric alternatives.



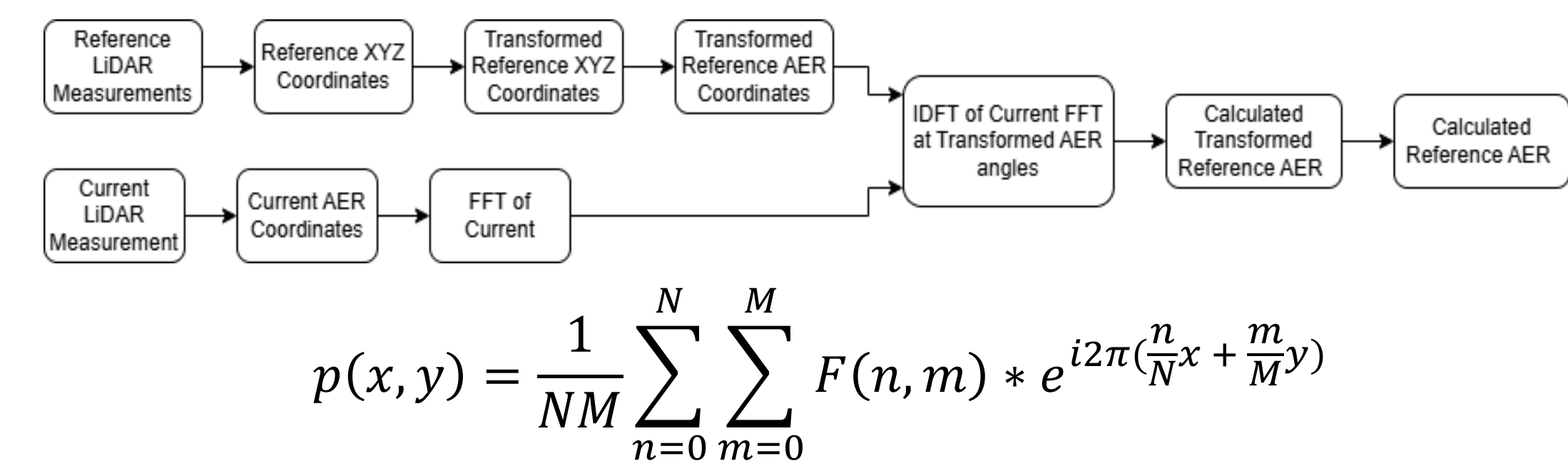
Contribution

This research develops a framework to evaluate six interpolation methods for LiDAR image interpolation, a critical step in frame-to-frame registration for comparing LiDAR frames. It demonstrates that methods implicitly suppressing high-frequency content generalize better across frames, while full-spectrum methods overfit to frame-specific discontinuity structure. Results suggest local geometric approaches outperform global spectral methods for this signal class.

Methods and Setup

Six interpolation methods were implemented and evaluated:

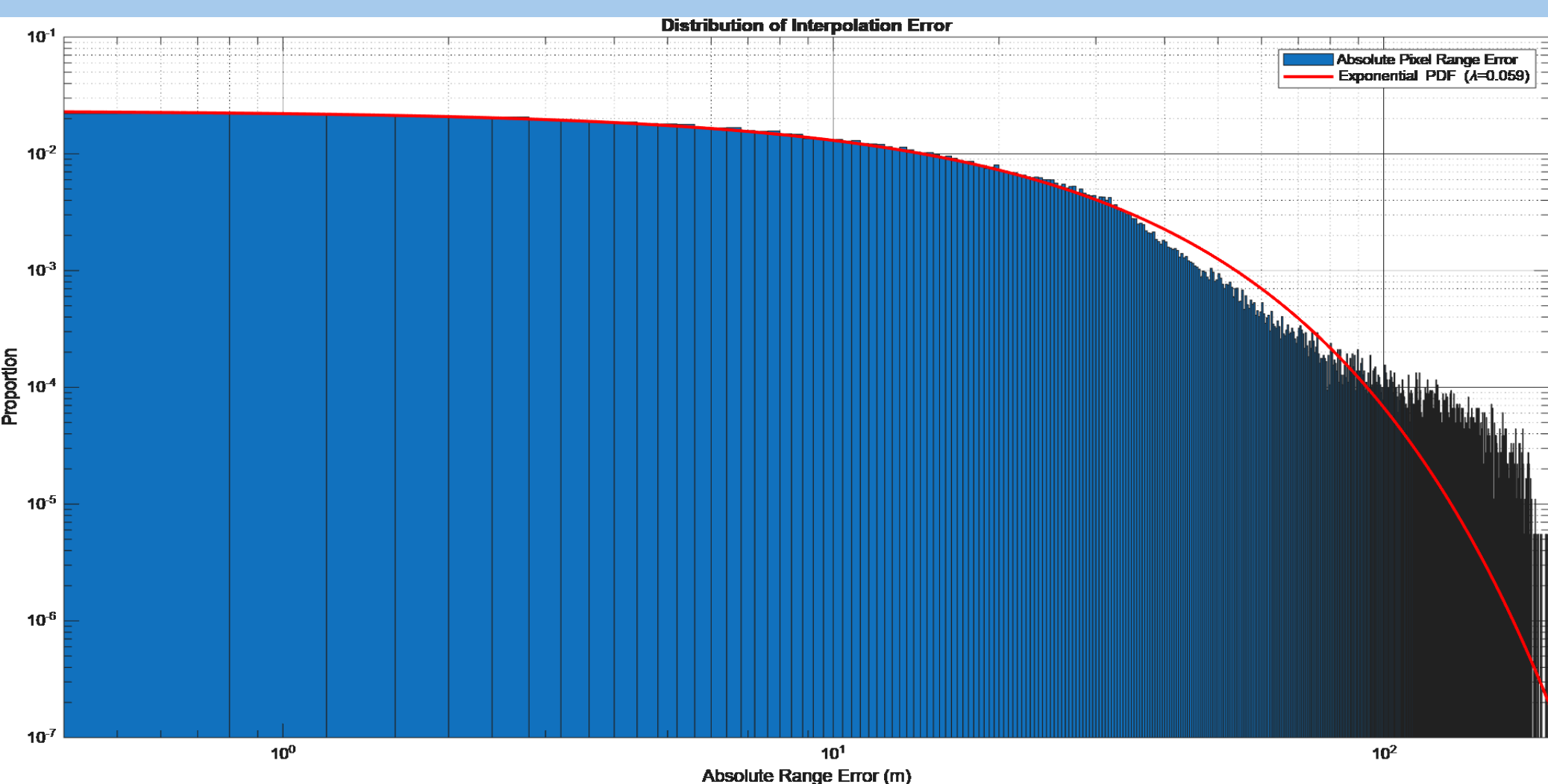
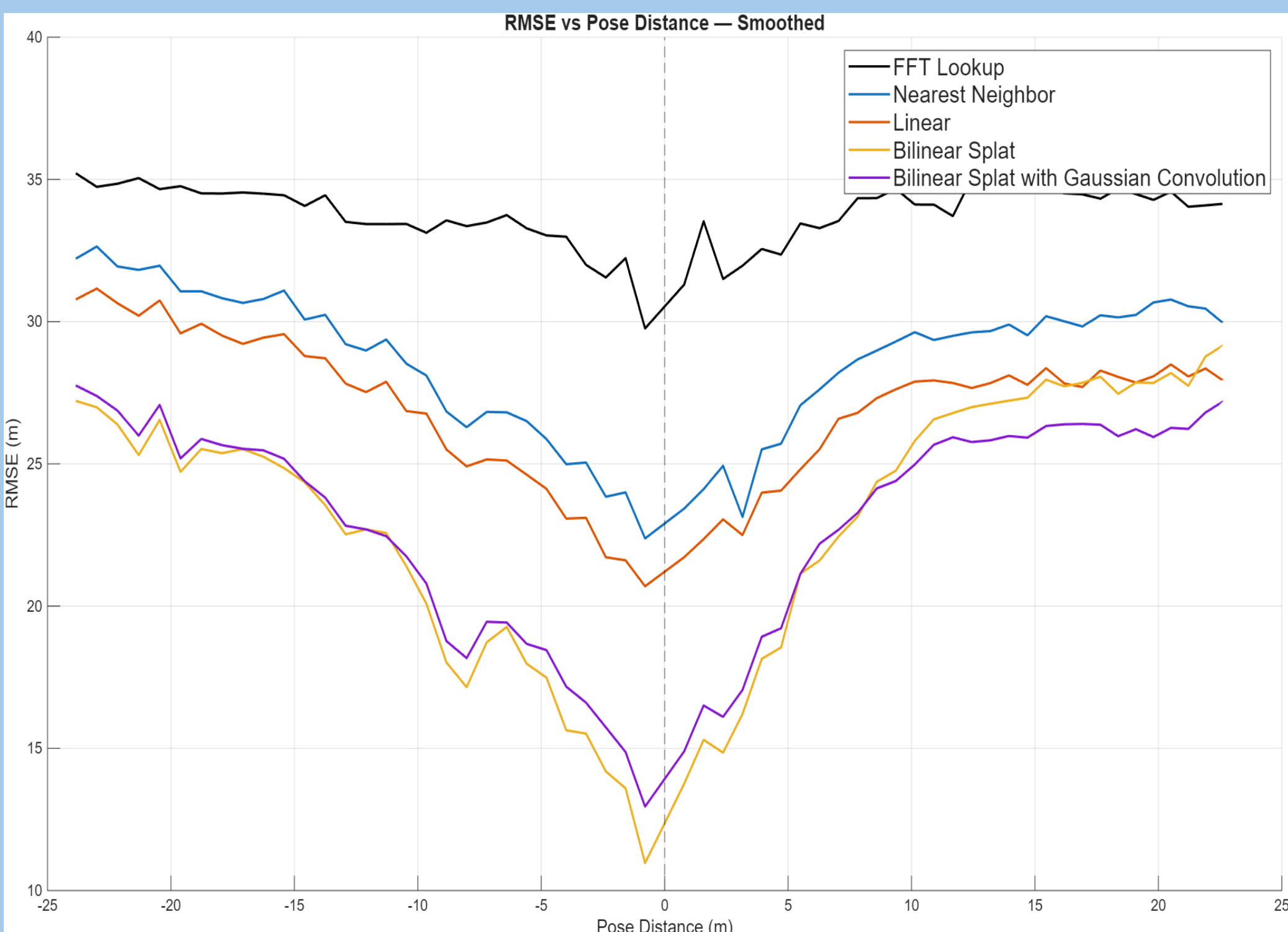
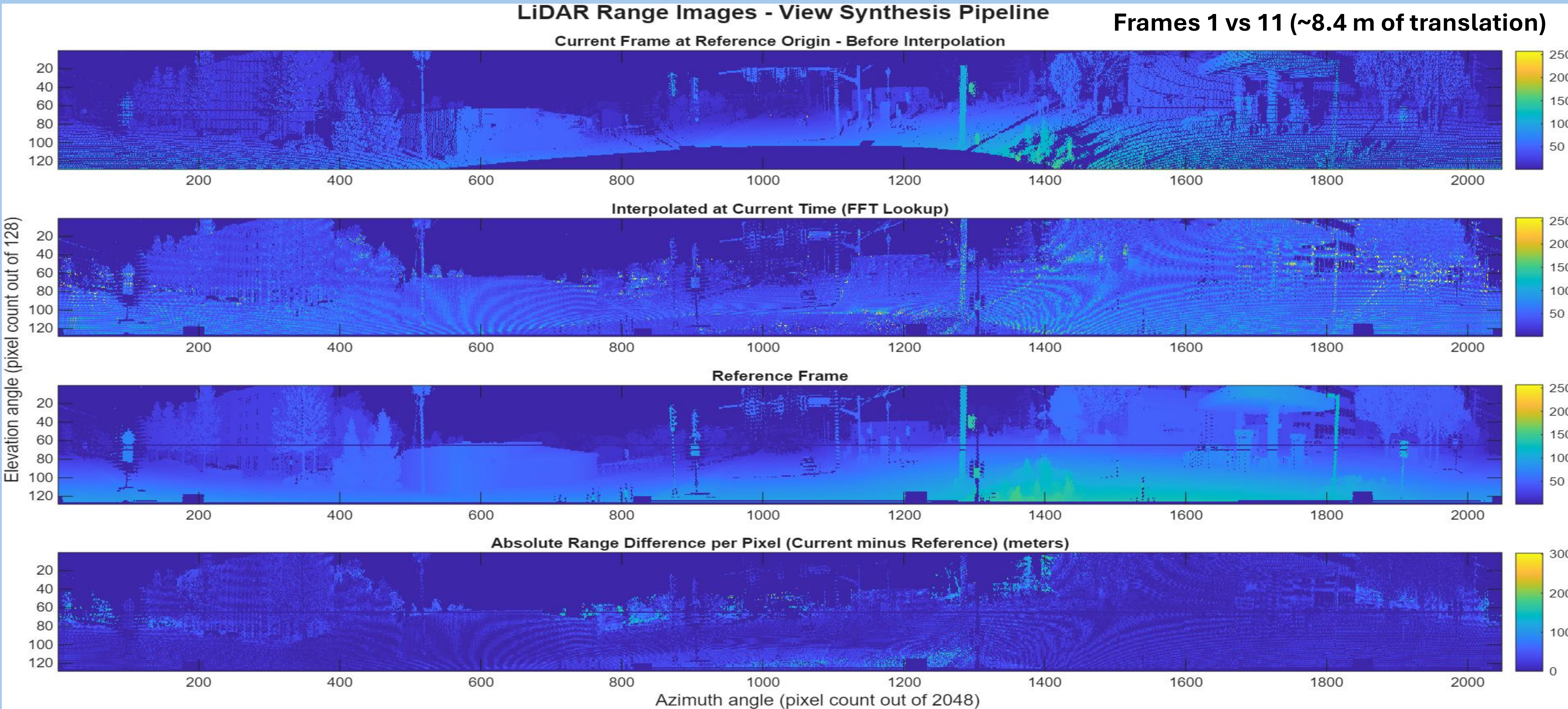
- Nearest Neighbor
- Linear
- Bilinear Splat
- Bilinear Splat with Gaussian Convolution
- FFT Lookup (Diagram Below)



Note: FFT Lookup has $O(N^2M^2)$ runtime, slower than other methods

To isolate interpolation error:

- Identical input: Same raw sensor measurements from a single Volpe Ouster OS2-128 scan from Fairfax, VA
- Registered using same ICP transformation.
- Frames have near-zero rotation to minimize distortion



Results

- Simple local methods outperform global across multiple tests
- RMSE decreases from NN to Linear to Splat as wider kernels suppress more high-frequency content, tracing an underfitting-to-optimal progression, while FFT Lookup's full-spectrum preservation produces significantly higher error
- Absolute error distribution (log-log scale) nearly matches exponential decay distribution, except for the deviations at the end of the tail.
- Large tail reveals a second distinct failure mode, likely at object boundaries
- Gibbs ringing has a global effect, with any depth discontinuity corrupting globally
- Note: Coverage decreases at large pose distances as fewer transformed points overlap with the reference grid, making direct pixel-wise comparison less reliable at the extremes

Conclusion

This work demonstrates that interpolation method has measurable impact on LiDAR frame comparison error, with global frequency-domain methods underperforming local geometric approaches. Global methods fail because high-frequency content in LiDAR range images is dominated by depth discontinuities, viewpoint-dependent features that do not transfer between frames, causing Gibbs ringing that corrupts the entire image. Local methods are more robust because finite spatial kernels act as implicit low-pass filters, attenuating frame-specific high-frequency content. For LiDAR signals, local spatial support is a more effective inductive bias than global spectral reconstruction, regardless of algorithm complexity. Future work includes characterizing each method's implicit frequency and evaluating downstream impact on pose estimation accuracy,

References:

Volpe National Transportation Systems Center. Ouster OS2-128 LiDAR Point Cloud Dataset, Fairfax VA, March 2023. U.S. Department of Transportation.